AlgoViz
Progress Service in REST
Java + Spring Boot

Submission cover and verification guide. Full answers are in assignment4_progress/NOTES.md; Maven source is included; build the runnable JAR locally.

Name	Roll number	Team responsibility
Anurag Tiwari	20251651028	Progress Service; Team Leader
Saurabh Singh	20251651089	Algorithm Service
Niraj Kumar Gupta	20251651066	Visualization Service
Mohd Rashid	20251651057	User Service

Scope and continuity

The submitted service implements the learning-progress part of Assignment 2's Progress Service. It is different from the Visualization Service used in Assignment 3. Visualization data is obtained through a separate HTTP service; ownership is preserved and no store is shared.

The CampusEats brief lists Catalogue, Orders or Delivery. This package explicitly adapts that category to AlgoViz Progress Service. Java 21 and Spring Boot align the implementation with the project stack; Java classes and JUnit serve as equivalents to the handout's Python modules and pytest. The equivalent classes and tests are mapped in NOTES.md. The intended group-repository layout places this service beside the Tutorial 4 folder, with earlier assignment artefacts retained separately.

Verified results

Gate	Actual result
OpenAPI validation	PASS - zero errors; OpenAPI 3.0.3; four operations
Four required tests	4 passed
Complete test suite	14 JUnit tests passed; zero failures or errors
Live HTTP evidence	Two Java processes; all 23 curl -i scenarios passed
Network behaviour	Timeouts, bounded retries, exponential backoff + jitter; no 4xx retry

Evidence provenance is recorded in evidence/environment.txt and PROVENANCE.md. The regeneration script refreshes these records together with the test and HTTP outputs on the machine where it runs.

No production deployment, authenticated access, frontend or live external SOAP gateway is claimed. The demonstration dependency serves seeded visualization records. The service and its idempotency storage are single-process and in memory.

Implementation & submission guide

REST resources

Method / resource	Behaviour	Success
POST /progress-records	Verify source session and create progress; Idempotency-Key required.	201 + Location
GET /progress-records/{id}	Read current public representation.	200
GET /progress-records?userId=28	Filter the collection; optional status filter.	200
PUT /progress-records/{id}/completion	Verify session completion before changing state.	200

All failures use application/problem+json with type, title, status and detail. The service demonstrates 400, 404, 409, 415, 422, 502 and 503; 413 guards body size and 500 is a sanitized fallback. No 202 is returned because processing finishes before the reply.

Evidence to review

File under assignment4_progress/evidence/	Proof
openapi-validation.txt	OpenAPI validator output
junit-output.txt	Maven build and all 14 JUnit tests, including four required cases
curl-transcript.txt	Actual curl -i commands, statuses, headers and bodies
network-checks.txt	Live scenario assertions passed
progress.log / visualization.log	Retry delays and received HTTP calls
environment.txt	Java 21, Maven and Spring Boot versions

Running and submitting

1. Install Java 21 and Maven; build and test with mvn clean package.
2. Run the locally built Spring Boot JAR and separate Java Visualization fixture. Optional Python utilities only regenerate evidence; they are not the backend.
3. Read NOTES.md for the resource table, design justification, fallback and all five comparison answers.
4. Repository layout: assignment4_progress beside Tutorial 4; demo and reference folders at the same parent level.
5. Tutorial 3 and Assignment 3 artefacts remain separate; submission uses the group repository workflow.

Assignment 3 comparison

The preserved WSDL has 100 lines; this OpenAPI contract has 324. NOTES.md quotes the actual ENGINE_UNAVAILABLE SOAP fault and compares it with the REST 503 problem representation, explicitly distinguishing the two services. All prior reference artefacts are retained without edits.

The stated deadline is before Lecture 6; the supplied brief gives no calendar date. This cover supplements the running service folder and NOTES.md; it does not replace the required source and evidence.